MINI PROJECT

Tic-Tac-Toe Game Using Python and Tkinter

Submitted By

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Project Guide

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Project Overview

This project is a graphical implementation of the classic Tic-Tac-Toe game developed using Python and Tkinter.

Objectives

- Develop an interactive desktop application using Python.
- Implement game logic for turn-based gameplay.
- Apply GUI development concepts using Tkinter.

Key Features

- Interactive 3×3 game board.

- Automatic winner detection.
- Draw condition detection.
- Reset functionality.

Technologies Used

- Python
- Tkinter

Learning Outcomes

- GUI development using Tkinter.
- Game logic implementation.
- Event-driven programming.

Execution

In [3]:

```
import tkinter as tk
from tkinter import messagebox

# Function to check winner
def check_winner():
    winning_combinations = [
        [0, 1, 2], [3, 4, 5], [6, 7, 8],
        [0, 3, 6], [1, 4, 7], [2, 5, 8],
        [0, 4, 8], [2, 4, 6]
    ]

    for combo in winning_combinations:
```

```python
        if buttons[combo[0]]["text"] == buttons[combo[1]]["text"] == buttons[combo[2]]["text"] != "":
            for i in combo:
                buttons[i].config(bg="lightgreen")

            messagebox.showinfo(
                "Winner",
                f"Player {buttons[combo[0]]['text']} Wins!"
            )
            return True

    # Check Draw
    if all(button["text"] != "" for button in buttons):
        messagebox.showinfo("Game Over", "It's a Draw!")
        return True

    return False


# Function when button is clicked
def button_click(index):
    global current_player

    if buttons[index]["text"] == "":
        buttons[index]["text"] = current_player

        if check_winner():
            return

        current_player = "O" if current_player == "X" else "X"
        status_label.config(text=f"Player {current_player}'s Turn")


# Reset Game
def reset_game():
    global current_player

    current_player = "X"

    for button in buttons:
        button.config(text="", bg="SystemButtonFace")

    status_label.config(text="Player X's Turn")
```

```python
# Main Window
root = tk.Tk()
root.title("Tic-Tac-Toe")
root.geometry("420x500")

current_player = "X"
buttons = []

# Heading
status_label = tk.Label(
    root,
    text="Player X's Turn",
    font=("Arial", 18)
)
status_label.pack(pady=10)

# Frame for Board
frame = tk.Frame(root)
frame.pack()

# Create 9 buttons
for i in range(9):
    button = tk.Button(
        frame,
        text="",
        font=("Arial", 24),
        width=5,
        height=2,
        command=lambda i=i: button_click(i)
    )

    button.grid(row=i // 3, column=i % 3)
    buttons.append(button)

# Reset Button
reset_btn = tk.Button(
    root,
    text="New Game",
    font=("Arial", 14),
    command=reset_game
```

```
)
reset_btn.pack(pady=20)

root.mainloop()
```

✅ Conclusion

The Tic-Tac-Toe Game project successfully demonstrates the implementation of game logic and graphical user interface development using Python and Tkinter. Through this project, concepts such as event handling, conditional statements, functions, and user interaction were applied in a practical way. The project provides an engaging and user-friendly gaming experience while strengthening programming, problem-solving, and GUI development skills.